

Reaction Time and the Distracted Driver

U01 / S01.4

Four deliberate departures from the locked lab template, all requested or forced by this lab. (1) No SAFETY section — requested 2026-09-22. There is no chemical, thermal, electrical or glassware hazard here; the only caution worth saying out loud is "nobody stands where a dropped stick can hit them," and that is in the procedure by implication rather than as a section. (2) No DISPOSAL / Alconox block — nothing is consumed and no glassware is used, so the standing block has nothing to act on. Replaced by a one-line reset step. (3) Data tables print as full grids rather than single-column copy-me skeletons, per Matt 2026-09-22. Students still record in the lab notebook — the printed grid is what they copy, and the handout still carries no answer blanks. (4) The student handout's body type is 10.1pt / 1.37 rather than the template's 10.3 / 1.40. This is the sanctioned page-budget lever (margins were not touched); it buys the fifth page back. This key is unchanged at the key template's 10.2 / 1.40.

PREP — FIVE MINUTES, MORNING OF

WHEN	TASK	DETAIL
Morning of	Count sticks	9 meter sticks for 27 students in groups of 3. Check each one for a readable 0 cm end — a stick with a worn or capped zero end gives a systematic error nobody will notice.
Morning of	Desk check	Groups need a flat front edge to rest a forearm on. Lab benches with a lip work; round tables do not.
First minute	Orientation check	Walk the room and confirm every group is holding the stick 0 cm down . This is the single error that silently ruins a whole group's data set.

Materials — 27 students, 9 groups, per period

Meter stick × 9 · calculator × 9 (or phones) · lab notebook and pen, one per student. Nothing consumable. Nothing to order.

TIMING — 50-MINUTE PERIOD

MIN	PHASE	WATCH FOR
0–5	Pre-lab check, roles assigned, sticks out	Pre-lab Q8 — did they actually draw three tables?
5–15	Part A — 3 catchers × 4 drops	Stick orientation. Fingers resting on the stick. Anticipation.
15–25	Part B — eyes closed	Callers watching A's hand instead of the stick. Misses being re-dropped instead of recorded.
25–34	Part C — counting backwards by sevens	Catchers who stop counting to catch. That trial is void.
34–47	Calculations — Working Your Data 1–5	cm → m conversion. This is where the period actually goes.
47–50	Sticks back, desks clear	

If a group is running behind, cut catchers, not situations. Have one designated catcher do all three situations rather than three catchers doing two. The comparison that matters is within one person — A vs B vs C on the same nervous system — so a group with one complete catcher has usable data and a group with three catchers and no Situation C does not.

FAILURE POINTS

- **The stick is upside down.** 100 cm at the bottom instead of 0. Every distance is then wrong by the length of the stick and the reaction times bunch up around **0.38–0.43 s** (a real 11–31 cm reads as 89–69 cm). Nobody reacts that uniformly, and that tell is the giveaway. Catch it in the first minute by walking the room — it is invisible in the data afterward.

- **Fingers already touching the stick.** The catcher feels the release instead of seeing it and posts times near 0.10 s. Enforce the 3 cm gap.
- **Anticipation.** Short outlier drops — under about 8 cm — are almost always a guessed release. Tell the dropper to vary the delay and tell the catcher to watch the 0 cm end of the stick, not the dropper's hand or face.
- **A Situation B miss gets re-dropped until they "get one."** This is the most damaging failure in the lab, because it deletes the result. A stick that falls past the hand means $t > 0.452$ s, and that is the finding, not a failed trial. Say this before Part B starts.
- **cm left as m.** 18.4 entered as meters gives $t = 1.94$ s. **Any reaction time over about 0.452 s that is not a recorded Situation B miss is a unit error** — 0.452 s is the ceiling a 100.0 cm stick imposes, so a caught stick cannot beat it. That single sentence catches most of the bad post-labs.
- **Square root of a negative.** Means they made Δy negative and a positive, or the reverse. Send them back to the sign convention they wrote in pre-lab Q4.
- **Counting backwards by sevens is genuinely hard for some students** and for anyone working in a second language. Counting back by threes, or reciting the alphabet backwards, loads attention the same way. Swap it without comment — the load is the variable, not the arithmetic.

EXPECTED RESULTS

Grade the method, not the number. Every student's value is legitimately their own, and the only hard bound is the meter stick itself — a caught stick cannot exceed 100.0 cm, so no measured reaction time can exceed **0.452 s**.

QUANTITY	ACCEPTED / EXPECTED	TYPICAL STUDENT RANGE
Situation A reaction time	0.20 s (supplied to students)	0.15–0.25 s · 11–31 cm. Published simple visual reaction time runs about 0.18–0.25 s.
Situation B reaction time	no published value — see note	PROVISIONAL: 0.30–0.45 s · 44–99 cm, with outright misses common.
Situation C reaction time	no published value — see note	PROVISIONAL: 0.22–0.35 s · 24–60 cm.
Spread across 4 good drops	—	1–4 cm is tight. Over about 8 cm means the procedure was not being run the same way twice.

Two things in that table are mine, not the literature's. The **0.20 s accepted value** is a published figure for simple visual reaction time in adults, where sources put the range at roughly 0.18–0.25 s; it is a population typical value, not a constant, and the student handout says so. The **Situation B and C ranges are my estimates**, converted from plausible reaction times — **no published value exists for these two conditions as run here**. Treat them as PROVISIONAL and replace them with your own first-period numbers, which will be better than anything I can give you.

SAFETY AND DISPOSAL — DELIBERATELY OMITTED

- No chemicals, no heat, no flame, no electricity, no glassware. Nothing to verify against an SDS and nothing to dispose of. The omission was requested, and the standing verification requirement does not attach because **the handout makes no safety or disposal claim at all** — which is the rule that matters. Never let a lab assert a safety fact it has not checked; asserting none is a different thing from asserting a wrong one.
- The one physical caution, if you want it said out loud rather than printed: a dropped stick that is missed lands on the floor, so keep feet back and do not run this standing on stools.

PRE-LAB KEY

- 1 Free fall definition. Gravity is the only force acting. Ignoring air resistance. Accept any answer naming the stick's density, its low speed, and the short drop distance as the reason drag is negligible. Do not accept "because we said so" or "air doesn't matter."
- 2 50 kg vs 90 kg student. The sticks land at the same time. The justification must reach acceleration: every object in free fall accelerates at 9.80 m/s^2 regardless of mass, and the mass of the person dropping it is irrelevant in the first place. Half credit for the right answer with a weight-based justification.
- 3 Why it measures reaction time. The stick is in free fall for exactly the interval between release and the fingers closing. The distance is a stand-in for that time. Grip strength changes how hard they squeeze, not when.
- 4 Sign convention. Down negative: $a = -9.80 \text{ m/s}^2$, Δy negative. Up-positive is the convention; accept down-positive if they then write $a = +9.80$ and Δy positive **and** use it consistently. The requirement is that it is declared, not which one they pick.
- 5 Part A measurement. Distance the stick fell · meter stick · to 0.1 cm. All three required.
- 6 Three void conditions. Chased the stick down · guessed the release and caught early · stick went past the hand. First two make the average look **faster** than reality; the third, if recorded as a short catch rather than a miss, also makes it look faster. Accept "slower" only for a genuinely recorded miss.
- 7 Student C's Part B job. Calling "GO!" the instant the stick is released — and watching **the stick itself**, not A's hand or face. The second half is the point of the question; it sets up post-lab Q2.
- 8 Three tables drawn. Full credit = three tables, in ink, units in the column headings, room to write. No credit for one table with the situations crammed in.

WORKING YOUR DATA — KEY

- 1 Averages. Four distances per person per situation, cm then m. Watch for groups averaging across *people* instead of across drops — that produces one number per situation and destroys the within-person comparison Q1 depends on.
- 2 Reaction times. $t = \sqrt{(2\Delta y \div a)}$, three sig figs, signs carried. Cross-check against the table in EXPECTED RESULTS above.
- 3 Percent error, Situation A only. $(|t_A - 0.20| \div 0.20) \times 100$. Worked example: $t = 0.194 \text{ s} \rightarrow (0.006 \div 0.20) \times 100 = \mathbf{3\%}$. **One sig fig** — the subtraction leaves 0.006, which has one. A student reporting 3.0% has claimed a precision the subtraction does not support; take it as a sig-fig error, not a wrong answer.
- 4 Spread. Highest minus lowest drop, Situation A, in cm. Typically 1–4 cm. This is the uncertainty they need for post-lab Q3(d).
- 5 Reset. Not graded. Checked.

POST-LAB KEY

Worked below with one illustrative student: **A = 0.19 s · B = 0.42 s · C = 0.30 s**. Every student's numbers differ. **Mark the method.**

- 1 Rank and compare. Expected ranking fastest to slowest: A, C, B. Difference $C - A = 0.30 - 0.19 = \mathbf{0.11 \text{ s}}$. Percent increase $= (0.11 \div 0.19) \times 100 = \mathbf{58\%}$. The sentence must land on **attention**, not vision — eyes were open and on the stick in both. Accept "my brain was busy," "I was doing two things," "I was processing the arithmetic." Do not accept "I couldn't see it."

2 The chain. Situation B contains **two** reaction times in series: Student C has to see the stick move and react by shouting, and only then can B react to the sound. The measured number is roughly C's visual reaction time plus B's auditory reaction time, so it is slow despite sound being the faster channel. The driving conclusion: a passenger's warning is worth less than it feels like, because it cannot reach you until the passenger has already reacted. **Full credit requires naming Student C's reaction time as part of the measurement.** A student who says "sound is slower" has not read the question and gets no credit for that clause.

3 Error analysis.

a. $\Delta y = -0.042 \text{ m}$, $a = -9.80 \text{ m/s}^2 \rightarrow t = \sqrt{(2(-0.042) \div (-9.80))} = \sqrt{(0.008571)} = \mathbf{0.093 \text{ s}}$ (2 sig figs — 4.2 cm has two).

Mark the signs. A student who drops them here after declaring the convention in pre-lab Q4 has not lost the arithmetic, but they have lost the habit the unit is teaching.

b. The catcher anticipated the release and squeezed before the stick moved — or was watching the dropper's hand rather than the stick. Either is acceptable and both are specific. **"Human error," "we messed up," and "the equipment was bad" earn zero**, per the standing rule.

c. With 4.2 included: mean = $65.5 \div 4 = 16.4 \text{ cm}$, so $\Delta y = -0.1638 \text{ m} \rightarrow t = \mathbf{0.183 \text{ s}}$. With 20.1 replacing it: mean = $81.4 \div 4 = 20.35 \text{ cm}$, $\Delta y = -0.2035 \text{ m} \rightarrow t = \mathbf{0.204 \text{ s}}$. The bad drop makes them look **0.021 s faster** than they are, about 10%. The honest average is 0.204 s, because the 4.2 cm drop did not measure a reaction — it measured a guess.

d. Spread of the three good drops = $21.1 - 19.8 = \mathbf{1.3 \text{ cm}}$, which is tight, so those three are trustworthy. Student compares against their own recorded spread. The reasoning to look for: a small spread means the procedure was repeatable; it does **not** mean the value is correct — a consistently upside-down stick has a tiny spread too.

4 The school zone.

a. $25 \text{ mi/h} \times (1609 \text{ m} \div 1 \text{ mi}) \times (1 \text{ h} \div 3600 \text{ s}) = \mathbf{11.2 \text{ m/s}}$ (11.17 unrounded).

b. $d = vt = 11.2 \times 0.19 = \mathbf{2.1 \text{ m}}$

c. $d = vt = 11.2 \times 0.30 = \mathbf{3.4 \text{ m}}$

d. Attentive: $2.1 + 8.9 = \mathbf{11.0 \text{ m}}$. Distracted: $3.4 + 8.9 = \mathbf{12.3 \text{ m}}$.

e. **1.2–1.3 m, and both are right.** Straight from the rounded totals, $12.3 - 11.0 = 1.3 \text{ m}$. Carried unrounded, $11.2 \times (0.30 - 0.19) = 1.23 \text{ m} \rightarrow 1.2 \text{ m}$. **Accept either** — but a student whose difference disagrees with their own parts (b)–(d) has an arithmetic error, not a rounding one. That is roughly 40% of the 3.0 m crosswalk; the child standing in the middle is 1.5 m from the near edge, so the extra distance is close to the whole margin between stopping at the edge of the crosswalk and arriving where the child is standing. Accept any one-sentence version that connects the number to the child. Do not accept a paragraph, and do not accept "you should pay attention" with no number in it.

The move students miss in Q4 is that $v = d \div t$ is legal for the reaction distance precisely *because* nothing is accelerating yet — the car is still at a constant 25 mph until the brake is touched. Students who reach for a kinematic equation there have not understood which phase is which. The handout's "push" box says this outright; if a student still misses it, the question to ask is "what is the acceleration during your reaction time?"

GRADING

Within the lab — PROVISIONAL, adjust to taste: pre-lab 8 · data and calculations 12 · post-lab 20 = 40 points. Post-lab Q4 is worth 8 of that 20; it is the assessment target of the whole activity. The carbonless copy is graded individually.

Gradebook category weight is not set for Physics. Chemistry's 55/20/15/7/3 split is explicitly *not* carried over — see [courses/physics/DECISIONS.md](#), open question 3. This lab is filed without a category weight rather than guessing one.

MISSED LAB

Do not re-run the setup. Hand the student a real class data table from this period — one complete group, all three situations — and have them work post-lab questions 1 through 4 from it. Add one question in place of the percent-error step they cannot do: *whose reaction time are you calculating, and name one thing you would have learned by catching the stick yourself that you cannot learn from this table.*